"PAPER_{0:D3}" -f [int]# PAPER #76 ■ Gamma-Ray Sources: Fermi-LAT + UQFF Emission Model

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---

2. UQFF Gamma-Ray Emission Model

Resonant Periodicity in Blazars

The UQFF Resonant mode predicts blazar gamma-ray flux modulation:

```
F_\gamma(t) = F_0 \times [1 + A_R \times \cos(\omega_{\rm blazar} t)]
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Where:

A_R = 10?5 (Resonant mode amplitude)

?_blazar = system spin/jet frequency (typically 10?7 to 10?5 rad/s for BL Lac objects)

For Markarian 421 (BL Lac, d=134 Mpc):

?_Mrk421 = 2p/(315 days) = 2.31■10?7 rad/s

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\Delta E_{\rm threshold} = m_{\gamma,\rm UQFF}^2 c^4 / (2 E_\gamma) \approx 10^{-60}
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Far below any observable threshold ? **Fermi-LAT spectral shapes are unmodified by UQFF.**

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3. Pulsar Gamma-Ray Phase Validation

For the Crab Pulsar (PSR J0534+2200, f = 29.65 Hz):

?_Crab = 2p ■ 29.65 = 186.3 rad/s

UQFF Resonant: g_R = cos(186.3t) ■ 10?5

At emission pulse (t = 0): g_R = 10?5 m/s■ maximum

This corresponds to a phase-dependent gravity variation of 3.6■10?8 relative to Newtonian g

The Fermi-LAT 4FGL catalog lists the Crab as **4FGL J0534.5+2201** with flux (100 MeV■100 GeV) = 5.65■10?7 ph/cm■/s. UQFF does not modify the average flux but predicts a 10?5 amplitude modulation component ■ below 4FGL timing precision for single-pulse analysis but potentially detectable in epoch-folded analysis.

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4. 4FGL Source Comparison Table

4FGL Source	Type	UQFF Prediction	Fermi-LAT Constraint
J0534.5+2201	Pulsar (Crab)	10?5 flux modulation	<1% in phase-folded
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J1653.9-0158	BL Lac (ultrafast)	Resonant peak	Not constrained

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Summary

Gamma-Ray Observable	Standard GR+QED	UQFF Prediction	4FGL Status
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Spectral shape	Power law/cutoff	Unmodified (m_? ~ 10?■5 eV)	Match
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Phase-pulse profile	Fixed	+10?5 at peak phase	Not constrained

Source: QCalcvalidation.py FERMILAT + FERMI_4FGL endpoints | ? = 0.0005/day | [SSq] = 0.57 .Groups[1].Value

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Abstract

The Fermi Gamma-ray Space Telescope 4th Source Catalog (4FGL-DR3, ~3800 sources, 100 MeV■1 TeV) provides the definitive census of gamma-ray emitters. The UQFF Resonant mode predicts periodic gamma-ray modulation in blazars and pulsars through the cos(?t) ■ 10?5 term. For high-energy gamma-ray emission, the UQFF also predicts a vacuum Cherenkov-like correction to the effective photon mass via the [UA] vacuum density. This paper validates UQFF emission predictions against 4FGL sources using the QCalc_validation.py Fermi endpoints.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0■10?4 day?■, [SSq] = 0.57) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

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UQFF computed: Canonical UQFF buoyancy parameter U*bi* = *?■[SSq]■GM/r■* = *5.0e-4■0.57■6.67e-11■M/r■; for solar parameters: U*bi,Sun = 5.7e-4■6.67e-11■1.99e30/(6.96e8)■ = 1.47e+2 m/s■.
